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PHY 101 — Past Questions & Materials
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Course: PHY 101
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 100

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Paper 1 of 2 — PHY 101 - 2022/2023 - First Semester - Physics Question Set

Level 100 • First Semester • PastQuestion

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PHYSICS QUESTION SET

◆ SECTION A: Multiple Choice Questions (MCQs)

1. Which coordinate system uses distance from origin and an angle?

- A. Cartesian
- B. Polar
- C. Rectangular
- D. Linear

Answer: B. Polar

2. In Cartesian coordinates, position is described using:

- A. r and θ
- B. Magnitude only
- C. x and y
- D. Direction only

Answer: C. x and y

3. Space is best described as:

- A. The region occupied by matter
- B. A form of energy

- C. A force
- D. A scalar quantity

Answer: A. The region occupied by matter

4. Time is classified as a:

- A. Derived quantity
- B. Vector quantity
- C. Fundamental quantity
- D. Dimensionless quantity

Answer: C. Fundamental quantity

5. Space–time is a concept that treats space and time as:

- A. Separate entities
- B. Independent quantities
- C. A single continuum
- D. Opposite quantities

Answer: C. A single continuum

6. Which of the following is a derived quantity?

- A. Length
- B. Mass
- C. Time
- D. Velocity

Answer: D. Velocity

7. A scalar quantity has:

- A. Direction only
- B. Magnitude only
- C. Both magnitude and direction
- D. Neither magnitude nor direction

Answer: B. Magnitude only

8. Which of the following is a vector quantity?

- A. Distance
- B. Speed
- C. Displacement
- D. Energy

Answer: C. Displacement

9. Vectors are commonly represented by:

- A. Dots
- B. Curves
- C. Arrows
- D. Numbers

Answer: C. Arrows

10. The length of a vector arrow represents its:

- A. Direction
- B. Origin
- C. Magnitude
- D. Sense

Answer: C. Magnitude

11. Scalar addition requires consideration of:

- A. Direction
- B. Angle
- C. Orientation
- D. Magnitude only

Answer: D. Magnitude only

12. Vector addition requires consideration of:

- A. Time
- B. Mass
- C. Direction
- D. Temperature

Answer: C. Direction

13. The triangle law of vector addition involves vectors placed:

- A. Tail to tail
- B. Head to head

C. Head to tail

D. Randomly

Answer: C. Head to tail

14. Mechanics is the branch of physics that deals with:

A. Heat

B. Light

C. Motion and forces

D. Electricity

Answer: C. Motion and forces

15. Classical mechanics applies mainly to:

A. Atomic particles

B. Subatomic particles

C. Macroscopic objects

D. Photons

Answer: C. Macroscopic objects

16. Quantum mechanics is mainly concerned with:

A. Planets

B. Vehicles

C. Atomic-scale systems

D. Buildings

Answer: C. Atomic-scale systems

17. Kinematics deals with:

A. Causes of motion

B. Effects of forces

C. Description of motion

D. Energy changes

Answer: C. Description of motion

18. Dynamics is the study of:

A. Motion without forces

B. Causes of motion

C. Graphs of motion

D. Time measurement

Answer: B. Causes of motion

19. Rotational motion occurs when a body:

A. Moves in a straight line

B. Spins about its own axis

C. Moves in a curve

D. Vibrates

Answer: B. Spins about its own axis

20. Circular motion occurs when a body:

- A. Rotates about its axis
- B. Moves around an external point
- C. Moves randomly
- D. Moves vertically

Answer: B. Moves around an external point

21. Angular displacement is measured in:

- A. metres
- B. seconds
- C. radians
- D. newtons

Answer: C. radians

22. Angular displacement refers to the change in:

- A. Linear distance
- B. Speed
- C. Angular position
- D. Acceleration

Answer: C. Angular position

23. Angular velocity describes the rate of change of:

- A. Speed

- B. Linear distance
- C. Angular displacement
- D. Force

Answer: C. Angular displacement

24. Angular acceleration is the rate of change of:

- A. Torque
- B. Angular velocity
- C. Linear velocity
- D. Momentum

Answer: B. Angular velocity

25. In circular motion, centripetal acceleration always acts:

- A. Tangentially
- B. Away from the centre
- C. Towards the centre
- D. Upward

Answer: C. Towards the centre

26. Centripetal force is responsible for:

- A. Increasing speed
- B. Changing direction
- C. Reducing mass

D. Producing torque

Answer: B. Changing direction

27. Torque is defined as the:

A. Rate of work

B. Turning effect of a force

C. Linear force

D. Energy per unit time

Answer: B. Turning effect of a force

28. Moment of inertia depends on:

A. Force applied

B. Shape and mass distribution

C. Speed

D. Temperature

Answer: B. Shape and mass distribution

29. Newton's first law is associated with:

A. Acceleration

B. Momentum

C. Inertia

D. Energy

Answer: C. Inertia

30. Newton's second law relates force to:

- A. Speed
- B. Velocity
- C. Acceleration
- D. Distance

Answer: C. Acceleration

31. Newton's third law states that:

- A. Force equals mass
- B. Action equals reaction
- C. Energy is conserved
- D. Momentum is zero

Answer: B. Action equals reaction

32. Momentum is the product of:

- A. Force and time
- B. Mass and velocity
- C. Energy and speed
- D. Acceleration and mass

Answer: B. Mass and velocity

33. Conservation of momentum applies when:

- A. External forces act
- B. The system is isolated

- C. Energy is lost
- D. Friction is present

Answer: B. The system is isolated

34. Work is said to be done when:

- A. A force is applied
- B. Energy is stored
- C. A force causes displacement
- D. Motion occurs

Answer: C. A force causes displacement

35. Work is a:

- A. Vector quantity
- B. Scalar quantity
- C. Fundamental quantity
- D. Dimensionless quantity

Answer: B. Scalar quantity

36. Energy is defined as the capacity to:

- A. Move
- B. Do work
- C. Apply force
- D. Produce heat

Answer: B. Do work

37. Kinetic energy is associated with:

- A. Position
- B. Motion
- C. Deformation
- D. Height

Answer: B. Motion

38. Potential energy depends on:

- A. Speed
- B. Motion
- C. Position or configuration
- D. Direction

Answer: C. Position or configuration

39. Power is defined as:

- A. Work done
- B. Energy stored
- C. Rate of doing work
- D. Force applied

Answer: C. Rate of doing work

40. Projectile motion occurs when an object moves under the influence of:

- A. Electric force only
 - B. Magnetic force only
 - C. Gravity only
 - D. Friction only
- Answer: C. Gravity only

41. The path followed by a projectile is called:
- A. Orbit
 - B. Trajectory
 - C. Displacement
 - D. Rotation
- Answer: B. Trajectory

42. In projectile motion, horizontal motion is:
- A. Accelerated
 - B. Uniform
 - C. Zero
 - D. Random
- Answer: B. Uniform

43. Elastic potential energy is stored in:
- A. Moving objects
 - B. Heated objects
 - C. Deformed elastic bodies

D. Falling objects

Answer: C. Deformed elastic bodies

44. The spring constant represents:

A. Mass of spring

B. Stiffness of spring

C. Length of spring

D. Energy stored

Answer: B. Stiffness of spring

45. A conservative force is one for which work done is:

A. Path dependent

B. Path independent

C. Always negative

D. Always zero

Answer: B. Path independent

46. In a conservative force field, work done in a closed path is:

A. Maximum

B. Minimum

C. Zero

D. Infinite

Answer: C. Zero

47. A non-conservative force:

- A. Conserves mechanical energy
- B. Has a potential energy function
- C. Depends on path taken
- D. Does zero work

Answer: C. Depends on path taken

48. Friction is an example of a:

- A. Conservative force
- B. Central force
- C. Non-conservative force
- D. Elastic force

Answer: C. Non-conservative force

49. Mechanical energy is the sum of:

- A. Work and power
- B. Force and distance
- C. Kinetic and potential energy
- D. Momentum and energy

Answer: C. Kinetic and potential energy

50. Vector resolution means breaking a vector

into:

- A. Smaller vectors
- B. Components along axes
- C. Scalars
- D. Forces

Answer: B. Components along axes

◆ SECTION B: True or False

51. Time is a fundamental physical quantity. — True

52. Velocity is a scalar quantity. — False

53. Displacement has both magnitude and direction. — True

54. Scalar quantities require direction for

addition. — False

55. Mechanics includes kinematics and dynamics. — True

56. Kinematics explains why motion occurs. — False

57. Rotational motion involves an axis within the body. — True

58. Circular motion requires centripetal force. — True

59. Angular displacement is a vector quantity. — True

60. Centripetal acceleration changes the

direction of velocity. — True

61. Torque depends on the line of action of force. — True

62. Moment of inertia depends only on mass. — False

63. Newton's first law is called the law of inertia. — True

64. Momentum is conserved in an isolated system. — True

65. Work is done even if there is no displacement. — False

66. Power measures how fast work is done. —

True

67. Projectile motion occurs in one dimension. —
False

68. Elastic potential energy exists due to
deformation. — True

69. Conservative forces have associated
potential energy. — True

70. Non-conservative forces dissipate mechani

Paper 2 of 2 — PHY 101 2012-2013 EXAM

Level 100 • First Semester • PastQuestion

Federal University of Petroleum Resources, Effurun, Nigeria

College of Science

Department of Physics

First Semester Exam (2012/2013 Academic Session)

Course Code: PHY 111 Time allowed: 2 hours

Name: OTERI AVWUNUDIIOGBA OPTION: VICK

Mat No:

Department:

Instructions: ANSWER ALL QUESTIONS (Take $g = 10\text{ms}^{-2}$)

- The SI Unit and dimension of Young Modulus is (a) Nm^{-2} and MLT^{-2} (b) $\text{kgms}^{-1}\text{s}^{-1}$ and $\text{MLT}^{-1}\text{T}^{-2}$ (c) Nm^{-2} and MLT^{-2} (d) N/m and MLT^{-4}
- Which of the following represents the SI unit of Viscosity (a) $\text{Kgm}^{-2}\text{s}^{-1}$ and MLT^{-2} (b) Nm^{-1} and LT^{-2} (c) kgms^{-1} and MT^{-2} (d) $\text{Kgm}^{-1}\text{s}^{-1}$ and $\text{ML}^{-1}\text{T}^{-1}$
- A physical quantity whose unit is the base unit is called (a) Foundation unit (b) Fundamental Unit (c) Derived Quantity (d) Ground Quantity
- The SI unit of Angular momentum is (a) $\text{Kgm}^2\text{s}^{-1}$ and ML^2T^{-1} (b) Nm^{-1} and MLT^{-2} (c) Kgms^{-1} and MT^{-2} (d) $\text{Kgm}^{-1}\text{s}^{-1}$ and $\text{ML}^{-1}\text{T}^{-2}$
- What is the SI unit and dimension of K in the equation : velocity = $k \times \text{density}$ (a) $\text{mkg}^{-3}\text{g}^{-1}$ and LMT^{-2} (b) $\text{mkg}^{-2}\text{s}^{-1}$ and $\text{LM}^{-2}\text{T}^{-1}$ (c) $\text{m}^4\text{kg}^{-1}\text{s}^{-1}$ and $\text{L}^4\text{M}^{-1}\text{T}^{-1}$ (d) None of the above
- The Period T of a vibration of a liquid surface drop depends on the density ρ , radius r and surface tension γ is given by $T = K\rho^x\gamma^y r^z$, What is x, y, z ? (a) 1, $\frac{1}{2}$, $-\frac{1}{2}$ (b) $-\frac{1}{2}$, $\frac{3}{2}$, $\frac{1}{2}$ (c) $\frac{1}{2}$, $\frac{1}{2}$, 1 (d) 0, 1, $\frac{1}{2}$
- Which of the following is not true about the dot product of two or three vectors (a) associative (b) commutative (c) orthogonal (d) acute
- Which of the following is not true of vector product? (a) $\mathbf{j} \cdot \mathbf{k} = 0$ (b) $\mathbf{i} \times \mathbf{i} = 0$ (c) $\mathbf{k} \times \mathbf{i} = \mathbf{j}$ (d) $\mathbf{k} \times \mathbf{k} = 1$
- Given that the mass of a moving object is 9kg and moving with a velocity $\mathbf{V} = 4\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, what is the momentum? (a) $1\mathbf{i} + 0.5\mathbf{j} + 0.67\mathbf{k}$ (b) $36\mathbf{i} + 27\mathbf{j} + 18\mathbf{k}$ (c) $24\mathbf{i} + 12\mathbf{j} + 16\mathbf{k}$ (d) $36\mathbf{i} + 9\mathbf{j} + 16\mathbf{k}$
- What is the instantaneous speed of a car moving according to the equation $x = 5t^2 + 3t + 12$ at a time of 3 seconds (a) 33m/s (b) 23m/s (c) 20m/s (d) 25m/s
- Given two vectors $\mathbf{A} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = 2\mathbf{j} + 3\mathbf{k}$, what is $\mathbf{B} \times \mathbf{A}$ (a) $5\mathbf{i} - 6\mathbf{j} - 5\mathbf{k}$ (b) $3\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$ (c) $3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$ (d) $-5\mathbf{i} + 5\mathbf{j} + 5\mathbf{k}$
- A car running at 70m/s accelerate at 22m/s^2 for 20seconds. What will be the final velocity of the car? (a) 30m/s (b) 50m/s (c) 70m/s (d) 110m/s

13. The motion of a particle whose velocity changes continuously in direction but not in magnitude is known as (a) Rectilinear motion (b) random motion (c) Uniform motion (d) oscillatory motion
14. if a number of forces acting at a point are in equilibrium, they can be represented in magnitude and direction by the number of sides taking in order is (a) Lami's theorem (b) triangle of forces (c) Polygon of forces (d) All of the above
15. A stone is thrown up vertically from the top of a cliff with an initial speed of 20m/s. If the height of the cliff above the ground is 30m, what is the maximum height reached? (a) 40m (b) 60m (c) 50m (d) 20m
16. A car approaching a bend in a road has its speed decreased from 30m/s to 10m/s while traveling a distance of 30m. What is the retardation of the car? (a) 13.33m/s² (b) 18.2m/s² (c) 2.3m/s² (d) 14.5m/s²
17. The speed that has a constantly changing acceleration directed towards the centre is called (a) uniform acceleration (b) centrifugal acceleration (c) non-uniform acceleration (d) centripetal acceleration
18. A ball on the edge of a vertical cliff is kicked horizontally with an initial velocity of 4.0m/s. If the cliff is 200m high. What is the time taken for the ball to arrive at the beach? (a) 6.32secs (b) 4.6secs (c) 8.42secs (d) 12.4secs
19. A particle moves in the horizontal direction from 0 with an initial velocity of 6m/s. At a time t seconds, later its acceleration is given by $a = 2t + 1$. Find the velocity after 4secs (a) 23m/s (b) 20m/s (c) 50m/s (d) 32m/s
20. Two masses 0.85kg and 0.6kg are suspended by means of an inextensible string which passes over a frictionless pulley. Find the acceleration of the two masses (a) 12.03m/s² (b) 1.72 m/s² (c) 7.03m/s² (d) 4.51m/s²
21. Which of the following correctly gives the relationship between linear acceleration and linear speed of a body moving uniformly in a circle of radius r ? (a) $a = \omega/r$ (b) $V = \omega r$ (c) $a = v^2/r$ (d) $a = \omega^2 r$
22. A force of 200N pulls a sledge of 50kg and overcomes a constant frictional force of 40N. What is the acceleration of the sledge? (a) 4.2m/s² (b) 4.9m/s² (c) 3.2m/s² (d)
23. A body of mass 0.1kg moving with a velocity of 6.0m/s collides head on with another body of mass 0.2kg at rest. Calculate the common velocity, if they move off together (a) 1.8m/s (b) 2.0m/s (c) 2.8m/s (d) 3.0m/s
24. An automobile company declares the maximum power delivered by the engine of a manufactured car of mass 1200kg to be 90kw. What is the minimum time in which the car would accelerate from rest to 30m/s² after 100m? (a) 40secs (b) 32secs (c) 56secs (d) 10secs
25. An object of mass 16kg moves in a circular path of radius 8×10^{-2} m with a linear

velocity of 0.7m/s. What is the angular velocity? (a) 8.8rad/s (b) 12.0rad/s (c) 6.7rad/s (d) 5.6rad/s

26. A disc of mass 50kg and radius 20cm has a moment of inertia of 2.5kgm² about its axis. Calculate the total kinetic energy when rolling horizontally at 3.0m/s (a) 400J (b) 503J (c) 506J (d) 600J

27. Which of the below expressions define angular momentum of a body? (a) $l\alpha$ (b) $mr^2\omega$ (c) $mr^2\omega$ (d) all of the above

28. A pendulum bob of mass 6kg is swung round a student's head in a horizontal circle of radius 1.5m with a velocity of 8m/s. What is the moment of inertia and kinetic energy of the bob? (a) 11.0kgm² and 123J (b) 13.5kgm² and 192J (c) 21.5 and 123J (d) 16.4 kgm² and 212J

29. Given that $k =$

30. All planets move in an elliptical orbit having the sun as one focus is also known as Kepler's law of (a) Period (b) orbit (c) area (d) length

31. Which of the statement below is correct about escape velocity? (a) it is velocity possessed by a body under the influence of gravity (b) it is the velocity of a body that is equal to the earth's centripetal force (c) it is the velocity a body must possess before it can escape the earth's gravitational pull (d) it is the velocity a person will possess to escape from his enemies.

32. A motion whose acceleration is directly proportional to its displacement from its equilibrium position and always directed toward the fixed point is (a) Circular motion (b) damping motion (c) simple harmonic motion (d) uniform planetary motion

33. A 5kg mass is observed to oscillate with a frequency of 2Hz and amplitude 4.5cm. Suppose the oscillation is SHM and undamped, what is the maximum velocity of the body? (a) 1m/s (b) 2.4m/s (c) 0.57m/s (d) 0.89m/s

34. in a simple harmonic motion experiment, the acceleration and velocity of the pendulum is maximum at and Respectively (a) equilibrium and maximum amplitude (b) maximum and minimum kinetic energy (c) both at equilibrium position (d) maximum amplitude and equilibrium

35. A particle moving with SHM has an amplitude of 2cm has a frequency of 29Hz. What is the period of the oscillation? (a) 2.10sec (b) 0.05sec (c) 5.40sec (d) 1.03sec

36. Which of the following is not observed within the elastic deformation stage of a material (a) proportionality limit (b) elastic limit (c) tensile stress (d) yield point

37. Which of the following is not a characteristic of ductile material? (a) It has yielding point (b) pass through elastic and plastic deformation (c) change is visible before breaking (d) easily fracture

38. A 4kg mass is attached to the end of a vertical wire of length 30cm and diameter

- 0.66m and the extension is 0.60mm. What is the Young Modulus of the wire? (a) $1.17 \times 10^8 \text{Pa}$ (b) $5.85 \times 10^{11} \text{Pa}$ (c) $2.0 \times 10^{11} \text{Pa}$ (d) All of the above
39. A wire of length 4m is extended by 2.5mm by a load of 300N. What is the energy stored in the wire. (a) 0.23J (b) 0.38J (c) 0.65J (d) 0.89J
40. The effects of surface tension can be reduced by the following methods except (a) Lower the temperature of the liquid (b) adding impurity (c) increase the temperature of the liquid (d) stirring the surface of the liquid
41. The surface tension of soap solution is given to be $2.5 \times 10^{-2} \text{N/m}$. Calculate the excess pressure inside the bubble of the liquid of radius 1.0cm (a) 60Pa (b) 22Pa (c) 10Pa (d) 34Pa
42. The liquids with acute angles of contact are said to glass (a) dry (b) form droplet (c) wet (d) break
43. When the pressure on any part of a confined fluid is changed, the pressure on every other part of the fluid is also changed by the same amount. This is principle? (a) Pascal's (b) Newton's (c) Poiseuille's (d) Archimedes
44. The flow rates of fluid through a pipe depend on the following except (a) The viscosity of the fluid (b) the diameter of the pipe (c) the pressure difference between the ends (d) temperature
45. Which of the following is not true about factors affecting blood pressure (a) blood velocity and blood volume (b) elasticity of blood vessels (c) blood viscosity and cardiac output (d) osmosis and cardiac output
46. What is the pressure difference between two ends of hose pipe of length 36m and radius 0.8cm, if the flow rate of Benzene through it is 0.75 litres per second. Take viscosity of Benzene to be $0.65 \times 10^{-3} \text{Pas}$ (a) $3.5 \times 10^3 \text{Pa}$ (b) $1.1 \times 10^4 \text{Pa}$ (c) $2.3 \times 10^3 \text{Pa}$ (d) $2.4 \times 10^3 \text{Pa}$
47. A Particle travels at constant speed in a circle in a circle of radius 9.0m and completes one revolution in 3.0secs. What is the acceleration of the particle? (a) 40ms^{-2} (b) 20ms^{-2} (c) 14ms^{-2} (d) 19ms^{-2}
48. A bucket of mass 10kg is whirled round on the end of a 1.2m long string in a vertical circle. If the speed of the bucket is 6ms^{-1} . What is the minimum tension in the string? (a) 400N (b) 200N (c) 300N (d) 800N
50. Which of the following is not an example of horizontal circular motion? (a) wall of death (b) aircraft banking (c) Rotating liquid surface (d) None of the above
51. A flywheel of moment of inertia 0.3kgm^2 is mounted on a horizontal axis of radius 0.01m. Find angular acceleration, if a force of 40N is applied tangentially to the axle (a) 33rads^{-2} (b) 2.4rads^{-2} (c) 1.33rads^{-2} (d) 3.4rads^{-2}
52. Johannes Kepler's law of planetary motion is in agreement with view of the solar system (a) Ptolemaic (b) Tycho Brahe (c) Copernican (d) all of the above
53. If a ball rolls down a plane, It will possess i) Rotational Energy ii) Translational

Energy iii) Angular Energy iv) Transitional Energy (a) i and iv (b) iii only (c) i and ii (d) i only

54. The Kinetic Energy of a flywheel of moment of inertia 0.05 kgm^2 rotating at an angular speed of 12 rads^{-1} is (a) 3.6J (b) 0.3J (c) 1.7J (d) 2.5J

55. If the kinetic Energy of a uniform circular disc is 32.4J what will be its moment of inertia if its angular speed is 18 rads^{-1} (a) 4.0 kgm^2 (b) 0.2 kgms^2 (c) 5.8 kgm^2 (d) 1.4 kgm^2

56. An athlete of mass 65kg jumps on the ground from a height 5m high. What is the force on the athlete when he lands flexing his knees and stops in 0.2s? (a) 1000N (b) 3250N (c) 3100N (d) 13000N

57. A 150g bullet travelling at 30 ms^{-1} strikes and adsorbed by a 85kg object. What is the speed with which the object moves off? (a) 1.06 ms^{-1} (b) 0.05 ms^{-1} (c) 2.09 ms^{-1} (d) 3.0 ms^{-1}

58. While accelerating from a rest position. A 100kg car is subjected to constant force of $6.1 \times 10^3 \text{ N}$. Using the linear Impulse principle, determine the speed of the car 8s later (a) 48.8 ms^{-1} (b) 42.6 ms^{-1} (c) 56.4 ms^{-1} (d) 32.8 ms^{-1}

59. During an impact time lasting approximately $5 \times 10^{-4} \text{ s}$ a golfer exerts an average force of about 5000N on the golf ball. What linear impulse is delivered to the ball as a result? (a) 2.5Ns (b) 3Ns (c) 5.5Ns (d) 3.2Ns

60. A Vertical Spiral Spring is pulled a distance of 20cm below its equilibrium position. If the forces constant of the spring is 1000 Nm^{-1} . Find the work done by the applied force (a) 35J (b) 25J (c) 20J (d) 30J

61. A System moving along a path is acted upon by a force $f(x) = 3x^2 + 2x + 8$. Find the work done by the force if the system moves a distance of $x = 1.0 \text{ m}$ to $x = 2 \text{ m}$ (a) 19J (b) 12J (c) 20J (d) 30J

62. Find the energy required to lift 50kg of rice from a ship into a container 15m above the ship floor (a) 13KJ (b) 7.5KJ (c) 6.7KJ (d) 15KJ

63. A ball of total potential energy at a top of a hill 70m high, falls freely from its rest position. What is the velocity half way to the ground? (a) $26. \text{ ms}^{-1}$ (b) 12.4 ms^{-1} (c) 45 ms^{-1} (d) 25.3 ms^{-1}

64. Which of the following correctly gives the relationship between linear acceleration and linear speed of a body moving uniformly in a circle of radius r? (a) $a = \omega/r$ (b) $V = \omega r$ (c) $a = v^2/r$ (d) $a = \omega^2 r$

65. A nut shell of mass 0.9kg attached to a string of length 0.5m is whirled in a horizontal circle at a constant speed. If the tension of the string is 35N, find the maximum speed of the nut (a) 40 ms^{-1} (b) 23.1 ms^{-1} (c) 4.4 ms^{-1} (d) 5.7 ms^{-1}

66. A conical pendulum of length 1.2m and mass 1.6kg has a period of 2.0secs. If the speed of the object in the circular motion is 1.42 ms^{-1} . What is the angle made to the vertical? (a) 3.4° (b) 6° (c) 2.6° (d) 8°

67. A wheel of radius 1.8m undergoes a rotation of uniform angular acceleration of 10.5 rad/s^2 , if its angular velocity is 2 rad/s . Calculate the linear velocity at $t=2 \text{ s}$
(a) 23 m/s (b) 41.4 m/s (c) 24.3 m/s (d) 34.3 m/s
68. _____ theorem states that the moment of inertia of a body about an axis is equal to the moment of inertia about a parallel axis through the Centre of gravity of the body (a) law of inertia (b) Parallel axis theorem (c) radius of gyration (d) all of the above
69. When a force acts on a body at a distance from the axis of rotation, it produces a turning effect called (a) Torque (b) Radius of Gyration (c) Flywheel (d) All of the above
70. The maximum static frictional force is known as (a) Static Friction (b) limiting static friction (c) dynamic friction (d) kinetic friction